Correlation exercise

(after B Railsback, www.railsback.org)

This exercise introduces you to the concepts of correlation combining several stratigraphic techniques.
The dataset consists of 9 logged outcrops. Logs are shown with their approximate lateral distances. All logs end at the top where exposure stops, and no log shows the full stratigraphic succession. Each log records the vertical thickness of 4 different lithologies: sandstone, mudrock, limestone and chalk. Additional paleontological data have been collected. Several volcanic ash beds provide numerical (absolute) ages. Significant erosion surfaces were also recorded.

Task:

Correlate the 9 logs by combining the various data available to you.
In the work file, all logs can be moved vertically to aid in correlations, but you do not need to move them laterally to maintain their spacing.

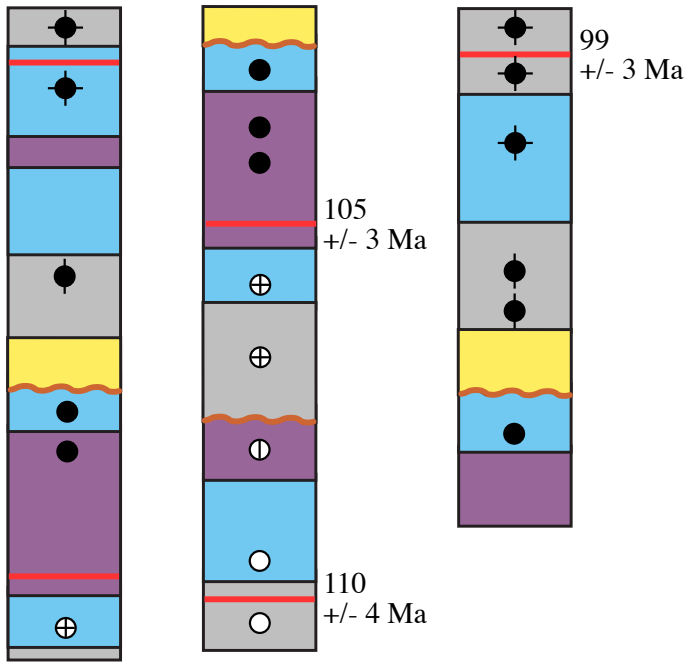
Procedure:

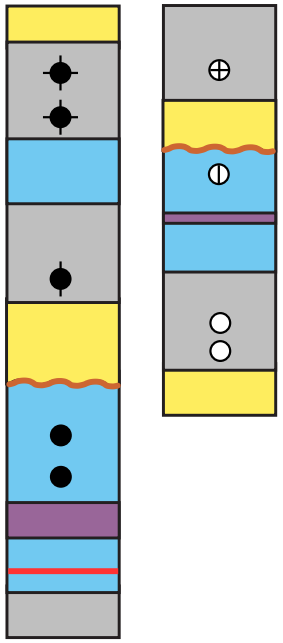
1. Use the paleontological data to establish biostratigraphic correlations.
2. Correlate the volcanic ash beds as marker beds.
3. Correlate the erosion surfaces.
4. Establish the lithostratigraphic correlations. It may be better to shade the areas of same lithology, rather than creating another set of correlation lines.
5. Break down the succession into ‘cycles’, or packages that show a consistent vertical succession of lithologies and the depositional environments they represent.

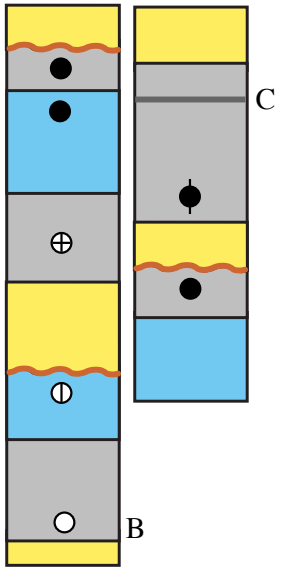
6. Answer the following questions:

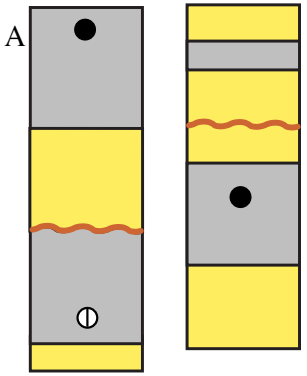
a. What geological process(es) may be responsible for the cycles identified in 5?
b. How old is the fossil labeled "A"?
c. How old is the fossil labeled "B"?
d. How old is the layer of rock labeled "C"?
e. Comparing the strata above the three chalks, why is there limestone and shale above the chalk in the uppermost cycle, just limestone above the chalk in the middle cycle, and neither limestone nor shale above chalk in the lowermost cycle?
f. Why is no fossil species found both below and above an unconformity?

Dataset









Sandstone (shoreline)

Mudrock (offshore)

Limestone (deeper offshore)

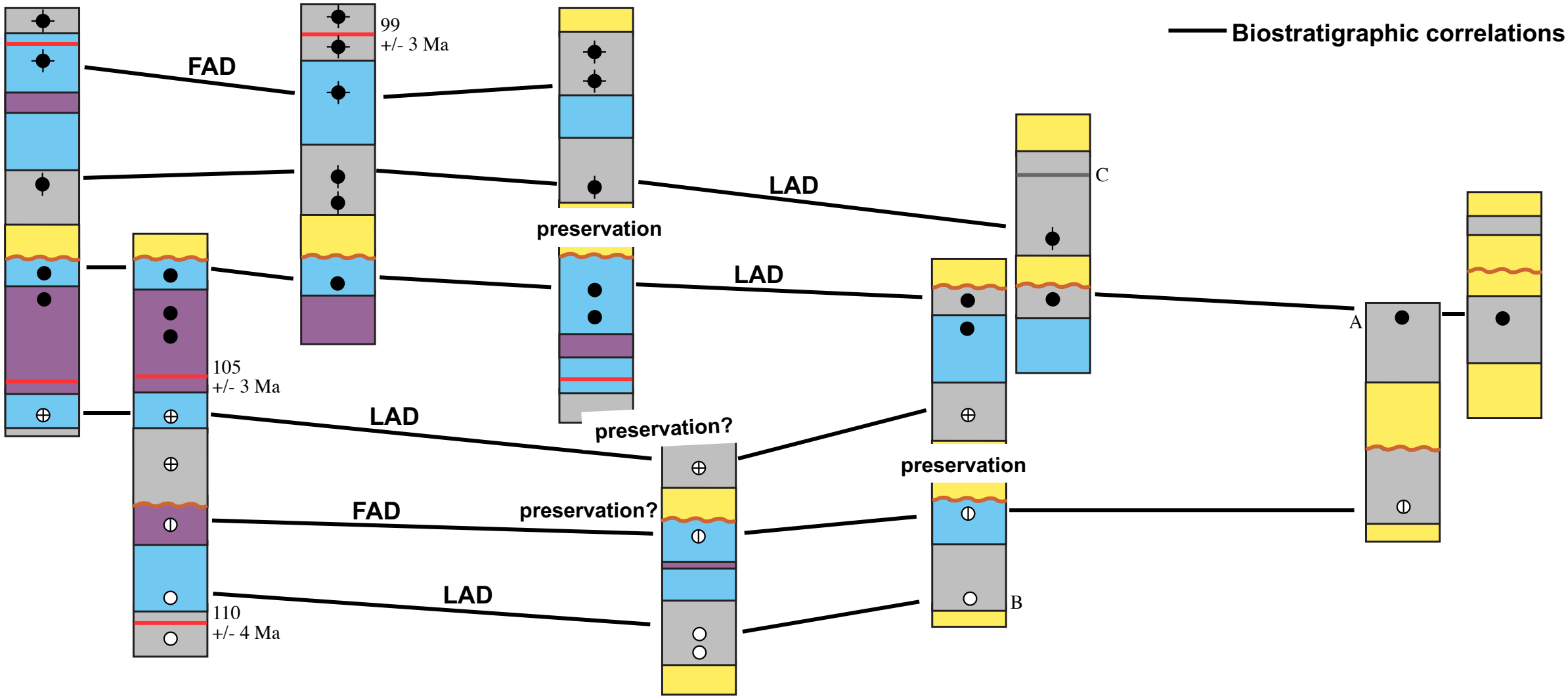
Chalk (deepest water)

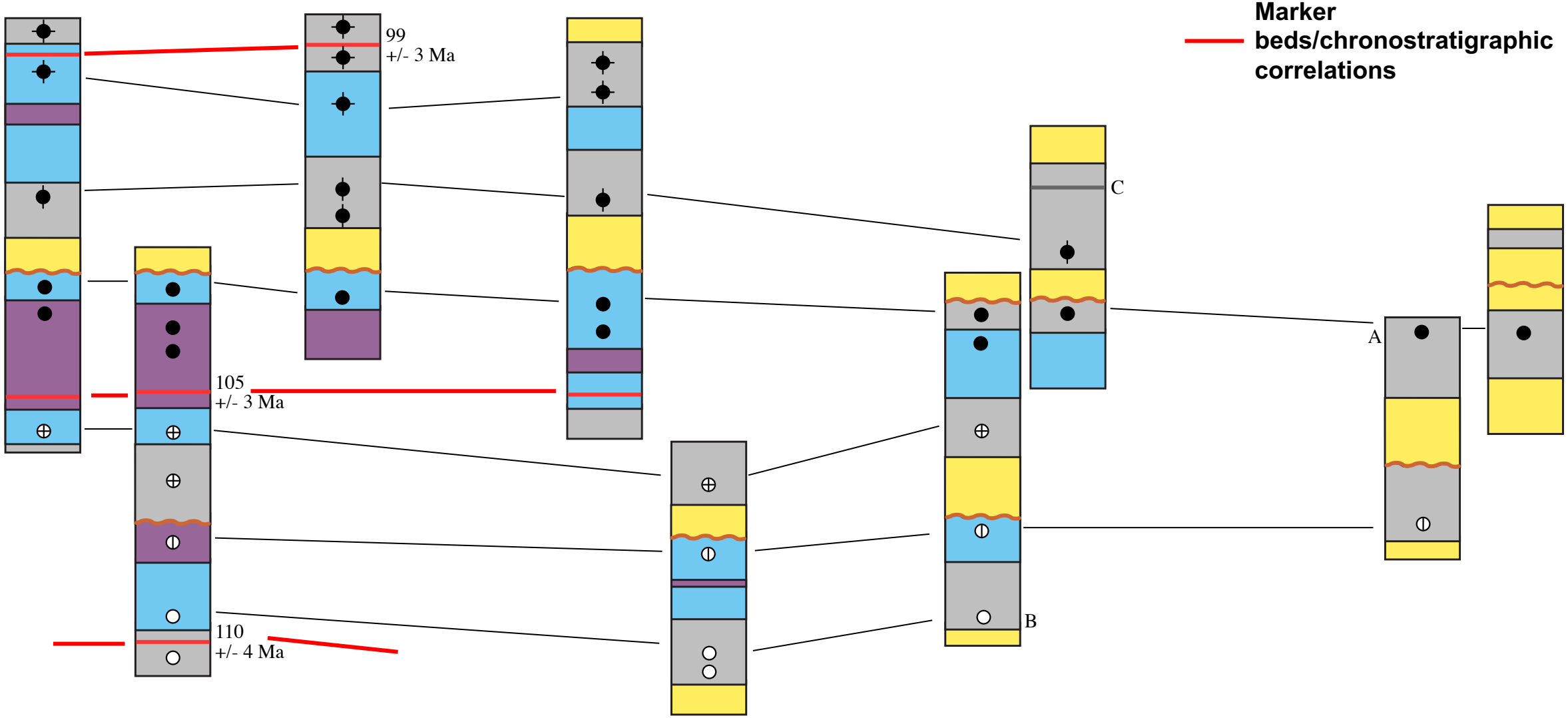
Fossils of 6 ammonite species

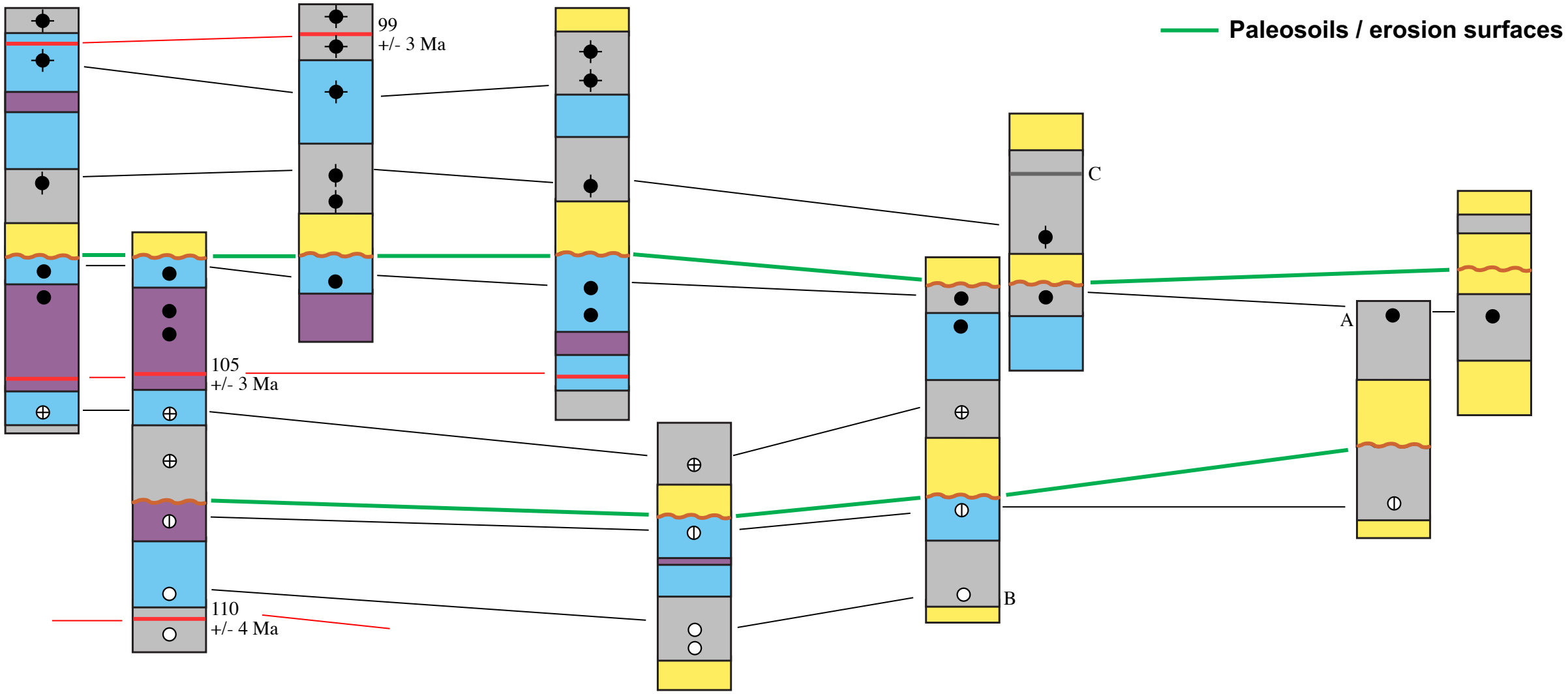
Volcanic ash

Paleosoil/erosion surface

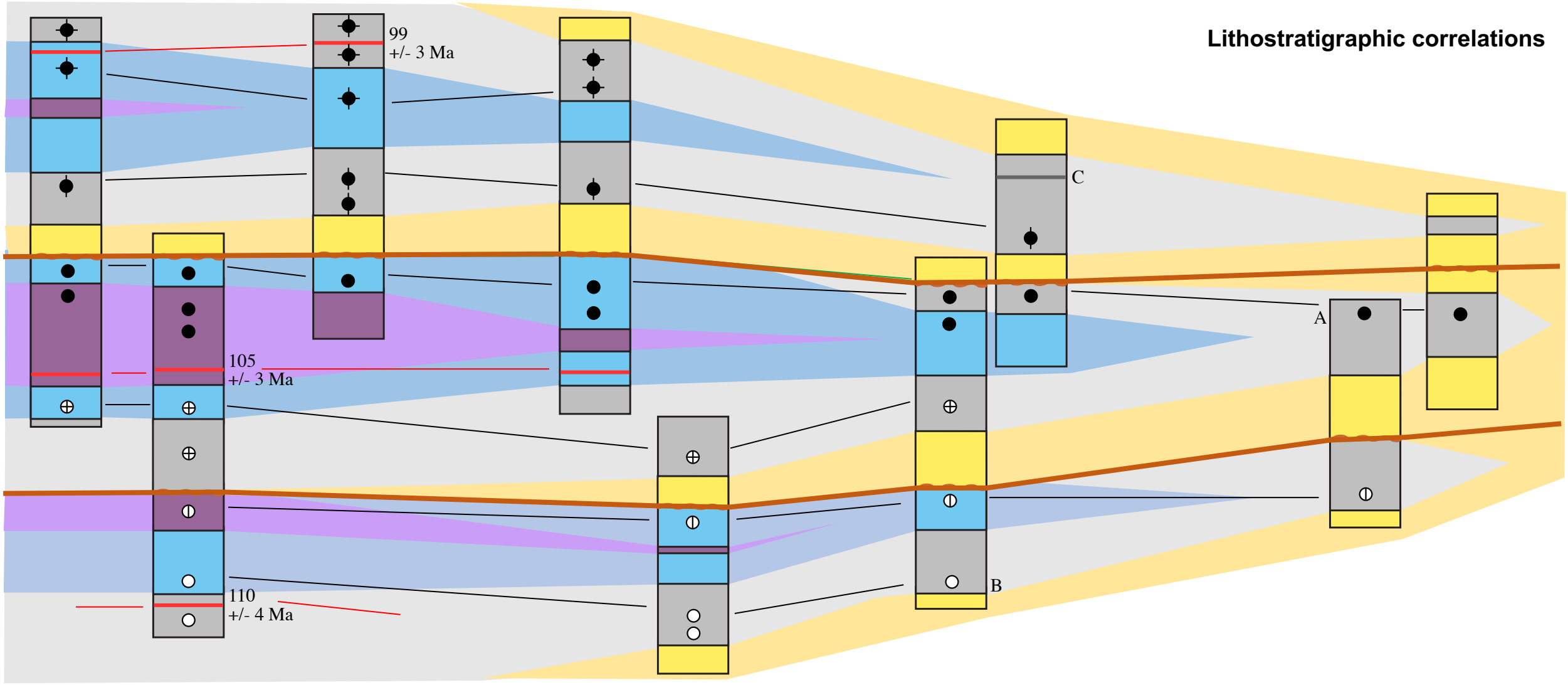
105 +/- 3 Ma Radiometric age

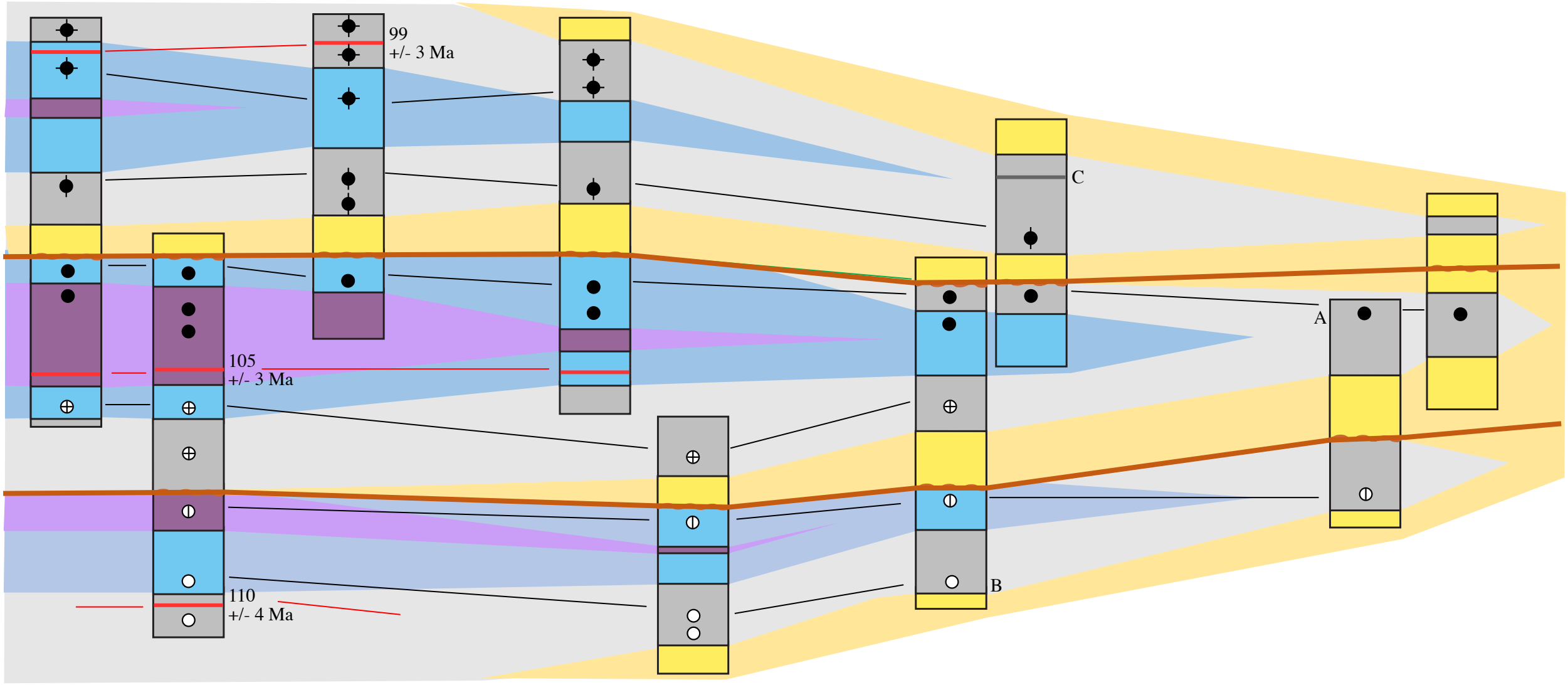

Biostratigraphic correlations
99
+/- 3 Ma
FAD
LAD
preservation
LAD
C
105
+/- 3 Ma
LAD
preservation?
preservation
A
FAD
preservation?
LAD
110
+/- 4 Ma
B


Marker beds/chronostratigraphic correlations
99
+/- 3 Ma
105
+/- 3 Ma
110
+/- 4 Ma
A
B
C


Paleosoils / erosion surfaces
99
+/- 3 Ma
105
+/- 3 Ma
110
+/- 4 Ma
C
A
B

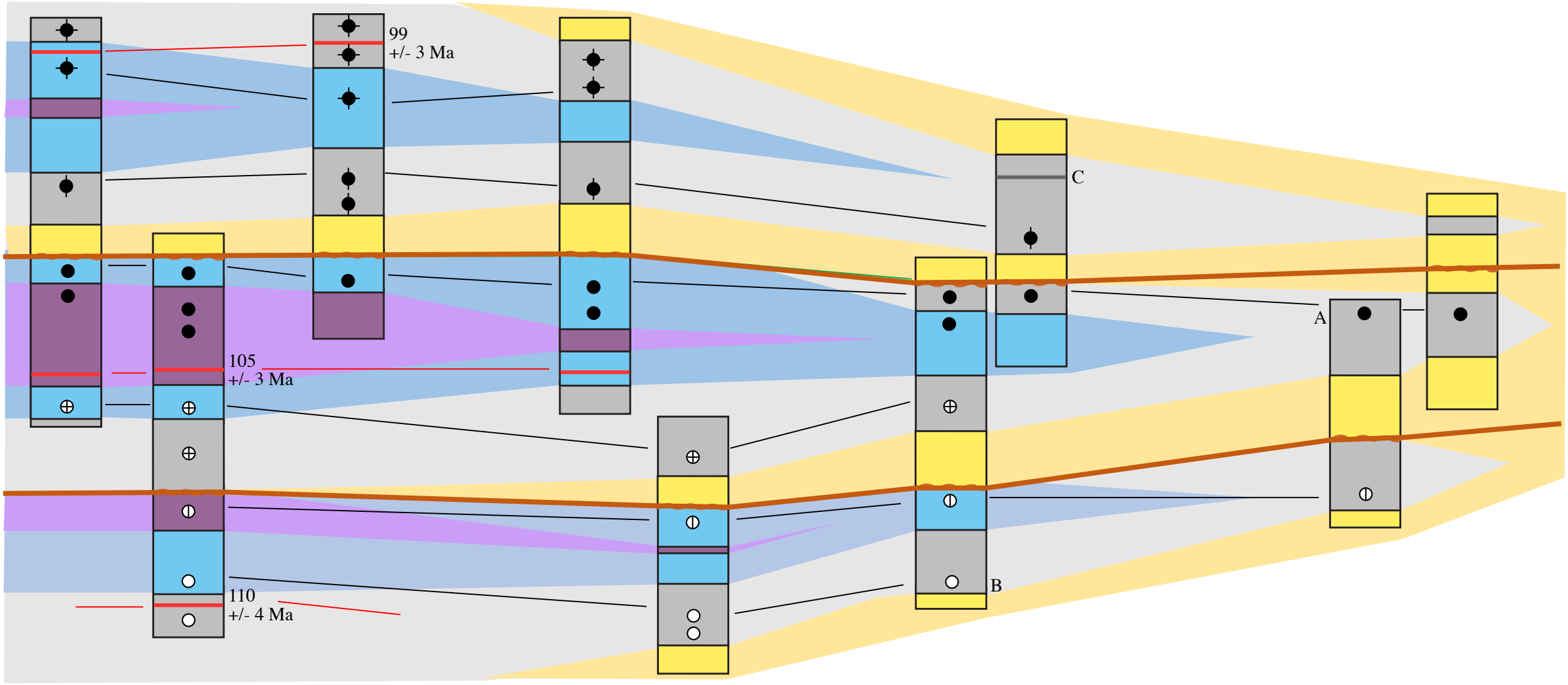
Lithostratigraphic correlations



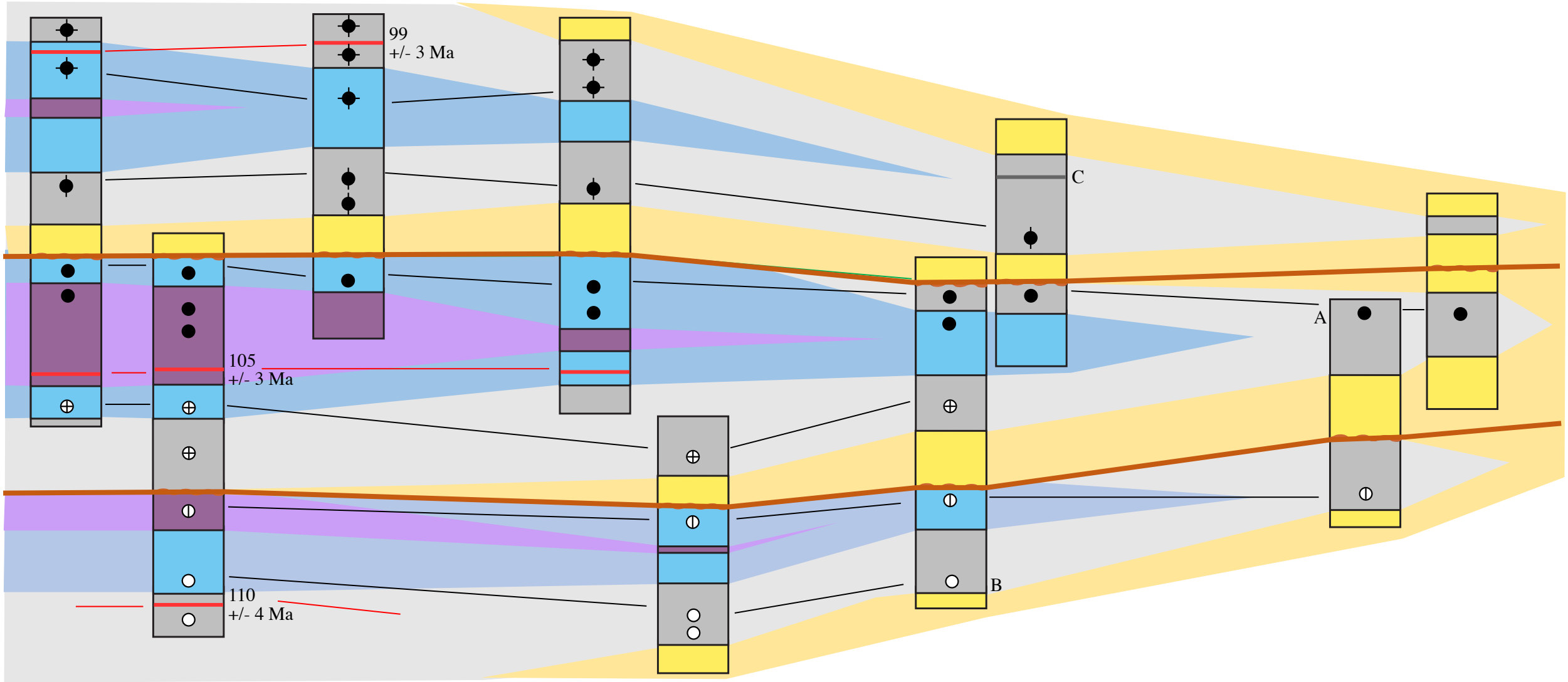


6a. What geological process(es) may be responsible for the cycles identified in 5?

relative sea level, transgressions and regressions; note that lithostratigraphic correlations cut time lines, i.e. they are diachronous



6b. How old is the fossil labeled "A"? ~103-104 Ma
6c. How old is the fossil labeled "B"? ~110 Ma
6d. How old is the layer of rock labeled "C"? ~100 Ma



6e. Comparing the strata above the three chalks, why is there limestone and shale above the chalk in the uppermost cycle, just limestone above the chalk in the middle cycle, and neither limestone nor shale above chalk in the lowermost cycle?

Erosion cut down more extensively during the fall of sea level between the 1. and 2. cycles, and then progressively less with each cycle

6f. Why is no fossil species found both below and above an unconformity?

missing time / missing record after erosion